

Deep Age Estimation Model Optimization

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Abstract— Face recognition is the predominant biometric method to identify and verify human facial features due to its stability in determining a person's identity. However, face recognition models can be improved relative to several challenges, e.g., the reliability of recognizing faces at different ages. Facial aging, which affects the shape and texture of the face, occurs in all individuals. Thus, this study attempted to optimize the performance of neural network-based age estimation methods by modifying the network architecture and hyperparameter values. In the future, we expect that the face recognition architecture will be employed to construct several models based on different age ranges. Consequently, the performance of the face recognition model within the specific age ranges is optimized. The modifications applied in this study resulted in a mean absolute error (MAE) value of 5.40 during validation for age estimation, indicating that the model estimated age more accurately than before, where MAE values of 7.60 and 5.53 were obtained.

Keywords— *Neural Network, Face Recognition, Face Age Estimation, Mean Absolute Error*

I. INTRODUCTION

Face recognition is a reliable biometric technology that employs artificial intelligence and computer vision technologies to identify and verify individual identities based on facial features [1]. Recently, face recognition has evolved rapidly and can be integrated seamlessly into various applications and systems [2], e.g., security systems [3], access control [4], forensics [5], and mobile devices [6]. However, several significant challenges remain in the development of face recognition models, e.g., expression variations [7], aging [8], and facial pose [9].

Expression changes, aging, and facial pose variations can be difficult to handle in the face recognition task [10], [11]. For example, facial deformations due to expressions and age-related wrinkles can alter appearances, as shown in Figure 1. Researchers have utilized deep learning methods and data augmentation techniques to address these challenges [12], and previous studies have demonstrated that face synthesis based on HyperStyle [13] as a data augmentation method with a focus on smile expressions and aging can enhance the performance of ResNet-50 face recognition models by 16% [14]. For example, when the ResNet-50 model was trained on the original FaceScrub dataset and a HyperStyle-synthesized dataset, an F1-score of 79% was obtained on the synthesized data, outperforming the F1-score of 63% obtained on the original FaceScrub dataset.

Recent research on pose variations has also enhanced the accuracy of SEResNet-18 models [15]. Following training with a merged dataset comprising the original dataset and a dataset with pose variations, the model achieved a validation

accuracy of 87%. In contrast, prior to this improvement, the SEResNet-18 model yielded a validation accuracy of 81% when trained on only the original FaceScrub dataset.

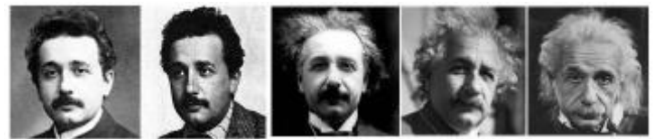


Fig. 1. Facial aging example [16]

Adjabi et al. [16] categorized face recognition into two main categories, i.e., face verification and face identification. In face verification, the system evaluates an image of an individual by comparing it to the face images in a database to determine whether the input image matches. In contrast, in face identification, the system identifies individuals by finding the best match among all faces in a registered model stored in the database. In other words, face identification performs one-to-all comparisons to determine individuals without providing prior identity declarations. The choice of architecture impacts the performance of a face recognition model substantially [17]. In deep-learning-based face recognition systems, the face recognition process, encompassing training and testing phases, is of paramount importance [18].

Numerous challenges, which represent prospects for further investigation, persist in face recognition, and one such challenge is reliable recognition of faces across various age groups [19]. Facial aging affects the facial shape and texture, including skin color and wrinkles, as shown in Figure 1. Age differences between the query face image and the reference face image of the same individual can reduce the accuracy of the face recognition system [20]. A practical application of recognizing faces at various ages is detecting specific individuals in a preregistered database, e.g., identifying missing individuals or suspects in a most wanted list in the law enforcement field [21].

Research related to age-based face recognition, which combines age-invariant face recognition (AIFR) and face age synthesis (FAS), known as multi-task learning face (MTLFace) [22], has demonstrated promising results. Experimental results obtained on the FG-NET dataset yielded an age prediction accuracy of 61.47% and a cosine similarity of 0.638 between the input and synthesized images. Another study on the age-invariant model (AIM) [23] achieved an accuracy of 84.81% on the cross-age face recognition dataset. In addition, research into age-invariant face recognition using convolutional neural networks (CNN) [24] obtained the highest rank-1 recognition percentage of

76.5% compared to other methods using the FG-NET dataset.

The primary contribution of this study is summarized as follows. We enhance the performance of neural network methods for age estimation [25] by modifying and adapting the model architecture. The main goal of this study was to confirm that the modified age estimation model outperforms the baseline model. In the future, we expect that this approach will contribute to face recognition model performance by estimating and categorizing the input images into a specific age range.

The remainder of this paper is organized as follows. In Section II, we describe related work, and the research methodology is described in detail in Section III. In Section IV, we discuss experimental results. Finally, the paper is concluded in Section V.

II. RELATED WORKS

A. General Face Recognition and Age-invariant Face Recognition

The inception of the general face recognition architecture can be traced back to the groundbreaking AlexNet, which won the 2012 ImageNet competition by introducing a CNN-driven method to handle the face recognition task. This renewed interest in the face recognition field, which began in the early 1990s [26]. Since then, CNN methods have been applied in many domains [1], and several architectures have utilized CNN techniques, e.g., GoogleNet [27], FaceNet [28], VGGNet [29], ResNet [30], and SENet [31].

The effectiveness of a face recognition system is impacted significantly by the neural network architecture used for model construction and the accessibility of face image training data [32]. Despite the notable achievements in facial recognition, the inherent age variations in facial images continue to pose a significant challenge in terms of realizing dependable face recognition systems [33]. Addressing age-related variations is challenging for face recognition systems attempting to identify individuals accurately in practical scenarios, e.g., locating missing children and identifying potential crime suspects [16]. Thus, AIFR is crucial for further development. According to the literature [22], the development of AIFR faces the following three main challenges.

- When there is a significant age difference, age variation can affect facial appearances, which leads to reduced face recognition performance.
- Age-based face synthesis is a complex process that involves aging and de-aging facial images (age progression and regression) because facial appearances change drastically over a long period and vary from one individual to another.
- Obtaining a large dataset of natural facial images to train models while preserving identity is nearly impossible.

To enhance image quality for generative models and offer interpretability for discriminative models, the MTLFace framework [22] handles AIFR and FAS concurrently. MTLFace comprises two primary tasks, i.e., acquiring representations related to identity and generating optimal facial synthesis. To elaborate further, MTLFace dissects

complex high-level features into two distinct and uncorrelated elements, i.e., those related to identity and those linked to age. These components are subsequently integrated within a multi-task learning framework. The age estimation task extracts age-related features, and the face recognition task (i.e., AIFR) is instrumental in removing identity-related parts.

B. Convolutional Neural Network (CNN)

Deep learning is a subset of machine learning situated within the domain of algorithms inspired by the architecture and functioning of artificial neural networks, which attempt to emulate human brain operations. This approach enables a computational model of multiple processing layers to discern data representations characterized by varying degrees of abstraction [34].

A CNN is specifically crafted to handle data in multiple arrays, e.g., a color image comprising three two-dimensional arrays containing pixel intensities for three color channels. A CNN is built on four key concepts that leverage the characteristics of natural signals, i.e., local connections, shared weights, pooling, and the utilization of multiple layers [34].

CNNs are highly effective multi-layered neural networks that are used to realize pattern recognition in digital images. The CNN architecture comprises two main components, i.e., feature extraction and classification. The feature extraction component includes a convolutional layer, a pooling layer, and an activation function, and the classification component encompasses one or more fully-connected layers [35]. The CNN architecture is illustrated in Figure 2.

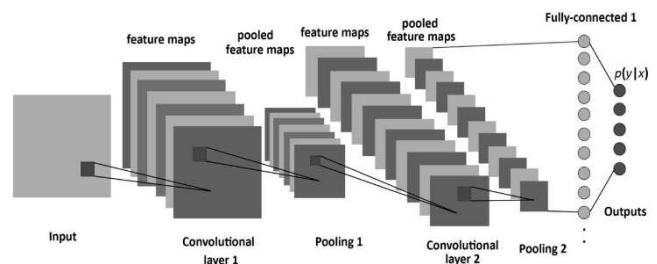


Fig. 2. CNN architecture [36]

ResNet is a renowned neural network architecture that won the ImageNet challenge in 2015. Its groundbreaking contribution lies in its ability to train deep neural networks effectively using more than 150 layers. Prior to the introduction of ResNet, training such deep networks presented a formidable challenge, primarily due to the gradient loss problem. The ResNet architecture shows that optimizing these neural networks becomes more manageable, thereby enabling them to achieve high accuracy even with significantly increased network depth [37]. The greater the depth of training, the more complex the network becomes. Note that depth is paramount in training because it allows parameters (or neurons) to retain and store optimal training values.

The ResNet architecture can be varied based on the number of layer configurations, including 18-layer, 34-layer, 50-layer, 101-layer, and 152-layer models. In Figure 3, the building blocks and their respective quantities are indicated within square brackets. In addition, the downsampling process, which attempts to reduce the output size, is executed

in the conv3_1, conv4_1, and conv5_1 layer with a stride of

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
		3×3 max pool, stride 2				
conv2_x	56×56	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$
conv3_x	28×28	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 8$
conv4_x	14×14	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 23$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 36$
conv5_x	7×7	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10 ⁹	3.6×10 ⁹	3.8×10 ⁹	7.6×10 ⁹	11.3×10 ⁹

2. These variations and processes are illustrated in Figure 3.

Fig. 3. ResNet architecture based on layer [37]

III. RESEARCH METHODOLOGY

A. Dataset

The experiments conducted in this study utilized the UTKFace benchmark dataset [38], which contains 23,705 human facial images annotated with three critical pieces of information, i.e., the age, gender, and ethnicity of the subjects.

An example of the UTKFace training and validation dataset randomly selected and used to construct the age estimation model is shown in Figure 4. The UTKFace dataset encompasses an age range spanning from 0–116 years. In addition, gender is indicated with 0 for males and 1 for females. There are also five ethnicity categories covered in the dataset, i.e., White, Black, Asian, Indian, and others (representing various ethnicities, e.g., Hispanic, Latino, and Middle Eastern).

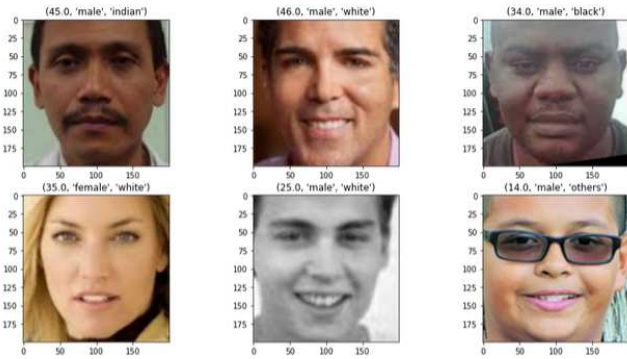


Fig. 4. Example of UTKFace dataset [38]

This dataset has become a crucial resource in face recognition and image processing research due to its diverse variations in age, gender, and ethnic background. It allows researchers to develop accurate and inclusive models and algorithms for face identification and attribute analysis. The UTKFace [38] dataset has been utilized in various scientific studies and practical applications, including the development of age recognition systems and facial demographic analysis. However, the age estimation experiments conducted in this study focused solely on age annotations.

In this study, the training and validation datasets were divided proportionally, with 80% of the data used for training and the remaining 20% allocated for validation. As a result, the number of images used to train the age estimation model was 18,964, and 4,741 images were used in the validation stage to refine the model. This approach ensures that the resulting model possesses strong generalization

capabilities to previously unseen data, which is a key factor when constructing effective and accurate models.

B. Experimental Stages

The experimental methodology used in this study was structured into four primary stages, as shown in Figure 5. The first stage involved the acquisition of a facial image dataset, which served as input data for the age estimation model. Here, we used the UTKFace dataset, as described previously. The second stage involved constructing an age estimation model using the baseline architecture of an existing neural network [25].

In the third stage, the age estimation model was trained using the first modified architecture of the neural networks. The neural network architecture was then refined in the fourth stage to minimize the mean absolute error (MAE).

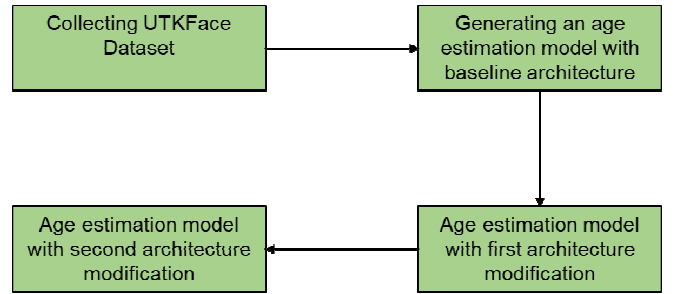


Fig. 5. Experimental stages

IV. RESULTS AND DISCUSSION

A. Experiments with the Original Age Estimation Model Architecture

These experiments used the original architecture without modifying the neural network layers. The hyperparameters were a batch size of 16, 100 epochs, a learning rate of 1e-4, and the Adam optimizer. The lowest MAE was observed at epoch 41, and the training time was approximately 50 minutes. Figure 6 (a) shows the training and validation MAE results.

B. Experiments with the First Modified Architecture of the Age Estimation Model

These experiments were conducted by adjusting the layers of the age estimation model's architecture. Here, several hyperparameter modifications were made to the original architecture, including a batch size of 64 and 300 epochs. Note that the learning rate remained at 1e-4, and the same Adam optimizer was used. The training and validation MAE results are shown in Figure 6 (b). The best MAE value was achieved at epoch 298, and the training time was approximately 2 hours and 8 minutes.

One architectural modification applied in this experiment was using LeakyReLU [39] as the activation function with parameter alpha value of 0.1. Note that LeakyReLU is an activation function similar to ReLU; however, it has a slight slope for negative values rather than being flat, which allows it to accommodate negative values that may occur during training. The slope coefficient is determined before training and remains constant throughout the training process. This type of activation function is prevalent in experiments that may encounter sparse gradients, e.g., training generative adversarial networks [40].

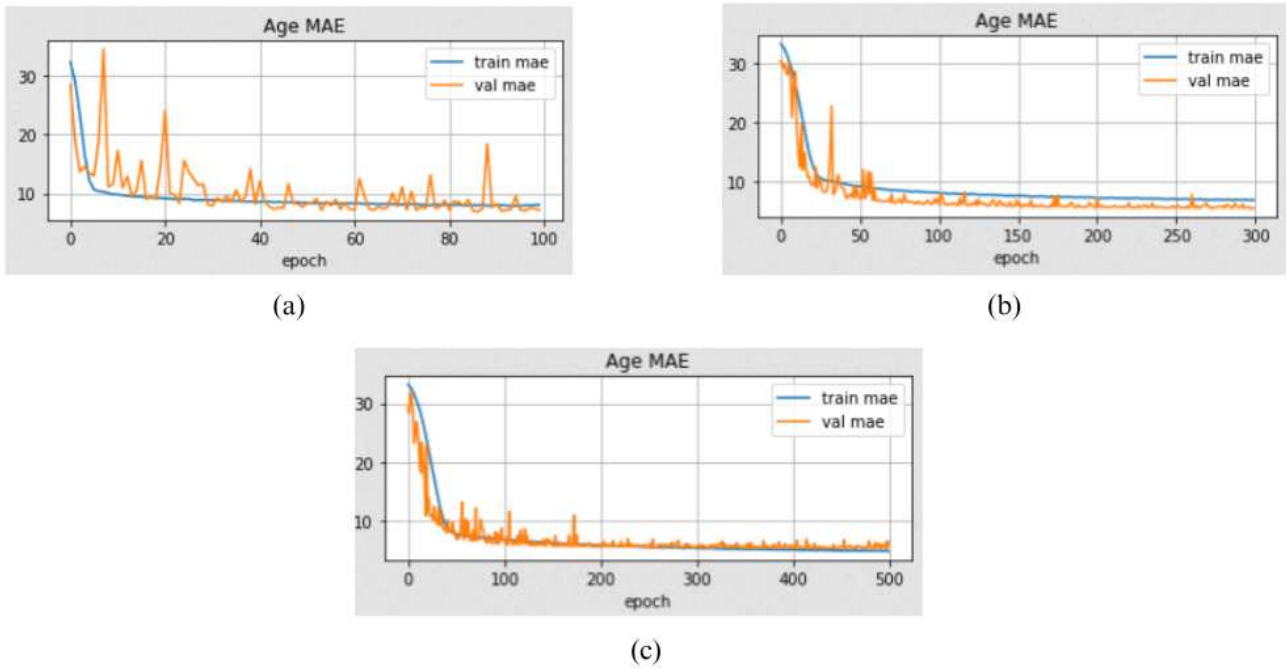


Fig. 6. Illustration of training and validation MAE diagram on age estimation model trained using (a) baseline architecture; (b) first architecture modification; (c) second architecture modification

In addition, several layers were added, and parameter values were adjusted. For example, dense layers were added, and their values were customized. The role of dense layers as fully-connected layers is to learn complex data patterns by combining features from all neurons in the previous layer. The output of the dense layers is a vector of values that can be used for classification and regression [41]. Dense layers can be combined with other layers, e.g., convolutional and pooling layers, to train more complex models. The number of neurons in a dense layer is a hyperparameter that can be adjusted to enhance model performance [42].

Furthermore, in these experiments, dropout layers were added and adjusted. Dropout layers act as a regularization technique for deep learning models to prevent overfitting, thereby ensuring that the model is not overly dependent on a single neuron. During training, the dropout layer randomly drops some neurons, which forces the remaining neurons to take on more or less responsibility for the input [43]. Dropout layers have improved performance consistently in various computer vision, speech recognition, and text classification tasks. Besides, dropout layers can also increase the speed of the computation process when training deep-learning-based models.

C. Experiments with the Second Modified Architecture of the Age Estimation Model

Here, the experimental process involved tuning the parameters of the architecture layers to train the model. In this experiment, the batch size was 128, 500 epochs were performed, the learning rate was set to $1e-4$, and the Adam optimizer was used. The highest accuracy was obtained at epoch 499, with training taking approximately 3 hours and 35 minutes. The corresponding training and validation MAE results are shown in Fig. 6 (c).

In this case, no additional layers were added or removed; however, in this experiment, the parameter values of the layer functions were adjusted, resulting in a more optimal age estimation model, as demonstrated by the low MAE values obtained during the model training and validation processes.

D. Evaluation of Age Estimation Model Training Experiments

Based on previous experiments, the optimal age estimation model was trained with modifications to the neural network layers. In addition, the hyperparameter configurations were adjusted to achieve maximum results, as indicated by age estimation's low MAE values. The best age estimation model was obtained in the second modification with a batch size of 128, 500 epochs, a learning rate of $1e-4$, and the Adam optimizer.

TABLE I. EXPERIMENTAL EVALUATION

No	Model	Epoch	Batch Size	MAE Training	MAE Validation
1	Model_V1 (original)	100	16	8.59	7.60
2	Model_V2	300	64	6.20	6.25
3	Model_V3	300	64	5.48	5.62
4	Model_V4 (Modified)	300	64	6.89	5.53
5	Model_V5	300	64	4.59	5.82
6	Model_V6	300	64	8.07	5.72
7	Model_V7	500	64	9.29	5.79
8	Model_V8	500	128	6.61	5.48
9	Model_V8 (Modified)	500	128	4.95	5.40

The best result occurred at epoch 499 with an MAE value for age estimation of 5.4 during validation. This result represents an improvement from the previous primary experiments with MAE values of 7.60 and 5.53, respectively. Table 1 compares the MAE values and the number of epochs and batch size used to train and validate each age estimation model. In the previous section, we only explained the results of the primary experiments, including the original architecture and the two modified architectures. These experiments are shown in bold in Table 1. However, Table 1 also summarizes other models resulting from the secondary experiments that contributed to constructing the optimal age estimation model.

V. CONCLUSION

The optimally generated age estimation model involves modifications applied to the neural network layers within the architecture, which was inspired by the findings of previous study. In addition, adjusting hyperparameter configurations plays a crucial role in achieving the best performance, as demonstrated by the low MAE values obtained in age estimation. The pinnacle of this achievement is found in the second modification, where the batch size was set to 128, 500 epochs were performed, the learning rate was set to $1e-4$, and the Adam optimizer was utilized. With this configuration, the most optimal results were obtained at epoch 499, with an MAE of approximately 5.4 during the validation phase.

In the future, the model should be trained on other benchmark datasets to compare results with what has been achieved. In addition, CNN architectures, e.g., VGGNet, FaceNet, and ResNet, should be investigated and compared in constructing face recognition models with a specific age range approach.

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REFERENCES

- [1] M. Wang and W. Deng, "Deep face recognition: A survey," *Neurocomputing*, vol. 429, pp. 215–244, 2021.
- [2] P. and P. I. and A. P. and C. D. Peng, "A face recognition software framework based on principal component analysis," *PLoS One*, vol. 16, no. 7, pp. 1–46, Oct. 2021, doi: 10.1371/journal.pone.0254965.
- [3] M. Baytamouny, R. Kolandaisamy, and G. S. ALDharhani, "AI-based Home Security System with Face Recognition," in 2022 6th International Conference on Trends in Electronics and Informatics (ICOEI), 2022, pp. 1038–1042. doi: 10.1109/ICOEI53556.2022.9776900.
- [4] Z. Wang, B. Huang, G. Wang, P. Yi, and K. Jiang, "Masked Face Recognition Dataset and Application," *IEEE Trans Biom Behav Identity Sci*, p. 1, 2023, doi: 10.1109/TBIOM.2023.3242085.
- [5] M. Taskiran, N. Kahraman, and C. E. Erdem, "Face recognition: Past, present and future (a review)," *Digit Signal Process*, vol. 106, p. 102809, 2020, doi: https://doi.org/10.1016/j.dsp.2020.102809.
- [6] D. Sunaryono, J. Siswanto, and R. Anggoro, "An android based course attendance system using face recognition," *Journal of King Saud University - Computer and Information Sciences*, vol. 33, no. 3, pp. 304–312, 2021, doi: https://doi.org/10.1016/j.jksuci.2019.01.006.
- [7] Y. Alaluf, O. Patashnik, and D. Cohen-Or, "ReStyle: A Residual-Based StyleGAN Encoder via Iterative Refinement," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021, pp. 6711–6720.
- [8] A. Othmani, A. R. Taleb, H. Abdelkawy, and A. Hadid, "Age estimation from faces using deep learning: A comparative analysis," *Computer Vision and Image Understanding*, vol. 196, p. 102961, 2020.
- [9] G. Guo and N. Zhang, "A survey on deep learning based face recognition," *Computer Vision and Image Understanding*, vol. 189, p. 102805, 2019, doi: https://doi.org/10.1016/j.cviu.2019.102805.
- [10] L. Boussaad and A. Boucetta, "Deep-learning based descriptors in application to aging problem in face recognition," *Journal of King Saud University - Computer and Information Sciences*, vol. 34, no. 6, Part A, pp. 2975–2981, 2022, doi: https://doi.org/10.1016/j.jksuci.2020.10.002.
- [11] I. Masi, S. Rawls, G. Medioni, and P. Natarajan, "Pose-aware face recognition in the wild," *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, vol. 2016-Decem, pp. 4838–4846, 2016, doi: 10.1109/CVPR.2016.523.
- [12] I. Masi, T. Hassner, A. T. Tran, and G. Medioni, "Rapid Synthesis of Massive Face Sets for Improved Face Recognition," *Proceedings - 12th IEEE International Conference on Automatic Face and Gesture Recognition (FG 2017)*, pp. 604–611, 2017.
- [13] Y. Alaluf, O. Tov, R. Mokady, R. Gal, and A. H. Bermano, "HyperStyle: StyleGAN Inversion with HyperNetworks for Real Image Editing," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 18511–18521.
- [14] M. Chaidir, T. F. Abidin, Hizir, and K. Muchtar, "HyperStyle-Based Data Augmentation to Improve the Performance of Face Recognition Model," in 2022 International Conference on Electrical Engineering and Informatics (ICEITICS), 2022, pp. 49–54. doi: 10.1109/ICEITICS56128.2022.9932083.
- [15] T. M. S. N. Alamsyah, T. F. Abidin, R. Ferdhiana, M. Dirhamsyah, and M. Chaidir, "Analysis of Face Data Augmentation in Various Poses for Face Recognition Model," in 2022 Seventh International Conference on Informatics and Computing (ICIC), 2022, pp. 1–6. doi: 10.1109/ICIC56845.2022.10006997.
- [16] I. Adjabi, A. Ouahabi, A. Benzaoui, and A. Taleb-Ahmed, "Past, Present, and Future of Face Recognition: A Review," *Electronics (Basel)*, vol. 9, no. 8, 2020, doi: 10.3390/electronics9081188.
- [17] P. Rahmatullah, T. F. Abidin, A. Misbullah, and others, "Effectiveness of Data Augmentation in Multi-class Face Recognition," in 2021 5th International Conference on Informatics and Computational Sciences (ICICoS), 2021, pp. 64–68.
- [18] X. Lv, M. Su, and Z. Wang, "Application of Face Recognition Method Under Deep Learning Algorithm in Embedded Systems," *Microprocess Microsyst*, p. 104034, 2021, doi: https://doi.org/10.1016/j.micpro.2021.104034.
- [19] M. K. Rusia and D. K. Singh, "A comprehensive survey on techniques to handle face identity threats: challenges and opportunities," *Multimed Tools Appl*, vol. 82, no. 2, pp. 1669–1748, 2023, doi: 10.1007/s11042-022-13248-6.
- [20] A. Kale and O. Altun, "Face age synthesis: A review on datasets, methods, and open research areas," *Pattern Recognit*, vol. 143, p. 109791, 2023, doi: https://doi.org/10.1016/j.patcog.2023.109791.
- [21] A. Dhamija and R. B. Dubey, "A novel active shape model-based DeepNeural network for age invariance face recognition," *J Vis Commun Image Represent*, vol. 82, p. 103393, 2022, doi: https://doi.org/10.1016/j.jvcir.2021.103393.
- [22] Z. Huang, J. Zhang, and H. Shan, "When age-invariant face recognition meets face age synthesis: A multi-task learning framework," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 7282–7291.
- [23] J. Zhao, S. Yan, and J. Feng, "Towards age-invariant face recognition," *IEEE Trans Pattern Anal Mach Intell*, vol. 44, no. 1, pp. 474–487, 2020.
- [24] M. Nimbarte and K. Bhoyar, "Age Invariant Face Recognition using Convolutional Neural Network," *International Journal of Electrical and Computer Engineering*, vol. 8, pp. 2126–2138, 2018, doi: 10.11591/ijece.v8i4.pp2126-2138.
- [25] W. Cao, V. Mirjalili, and S. Raschka, "Rank consistent ordinal regression for neural networks with application to age estimation,"

- Pattern Recognit Lett, vol. 140, pp. 325–331, 2020, doi: <https://doi.org/10.1016/j.patrec.2020.11.008>.
- [26] G. E. Krizhevsky, A. Sutskever, I. Hinton, “Imagenet classification with deep convolutional neural networks,” *Adv Neural Inf Process Syst*, pp. 1097–1105, 2012.
- [27] C. Szegedy et al., “Going Deeper with Convolutions,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 319–338. doi: 10.1109/CVPR.2015.7298594.
- [28] F. Schroff, D. Kalenichenko, and J. Philbin, “FaceNet: A Unified Embedding for Face Recognition and Clustering,” in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2015, pp. 815–823.
- [29] K. Simonyan and A. Zisserman, “Very deep convolutional networks for large-scale image recognition,” *3rd International Conference on Learning Representations, ICLR 2015 - Conference Track Proceedings*, pp. 1–14, 2015.
- [30] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, pp. 770–778, 2016.
- [31] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, “Squeeze-and-Excitation Networks,” *IEEE Trans Pattern Anal Mach Intell*, vol. 42, no. 8, pp. 2011–2023, 2020.
- [32] I. Masi, A. T. Tr  n, T. Hassner, J. T. Leksut, and G. Medioni, “Do we really need to collect millions of faces for effective face recognition?,” *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)*, vol. 9909 LNCS, pp. 579–596, 2016.
- [33] S. Singh and S. V. A. V. Prasad, “Techniques and challenges of face recognition: A critical review,” *Procedia Comput Sci*, vol. 143, pp. 536–543, 2018.
- [34] Y. Lecun, Y. Bengio, and G. Hinton, “Deep learning,” *Nature*, vol. 521, no. 7553, pp. 436–444, 2015, doi: 10.1038/nature14539.
- [35] M. Khoshdeli, R. Cong, and B. Parvin, “Detection of nuclei in H&E stained sections using convolutional neural networks,” *2017 IEEE EMBS International Conference on Biomedical and Health Informatics, BHI 2017*, pp. 105–108, 2017, doi: 10.1109/BHI.2017.7897216.
- [36] S. Albelwi and A. Mahmood, “A framework for designing the architectures of deep Convolutional Neural Networks,” *Entropy*, vol. 19, no. 6, 2017, doi: 10.3390/e19060242.
- [37] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” *Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition*, vol. 2016-Decem, pp. 770–778, 2016, doi: 10.1109/CVPR.2016.90.
- [38] S. Y. Zhang Zhifei and H. Qi, “Age Progression/Regression by Conditional Adversarial Autoencoder,” in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- [39] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, “Yolov4: Optimal speed and accuracy of object detection,” *arXiv preprint arXiv:2004.10934*, 2020.
- [40] T. Ridnik, H. Lawen, A. Noy, E. Ben Baruch, G. Sharir, and I. Friedman, “TResNet: High Performance GPU-Dedicated Architecture.” 2020.
- [41] Q. Zhang, Y. Liu, C. Gong, Y. Chen, and H. Yu, “Applications of deep learning for dense scenes analysis in agriculture: A review,” *Sensors*, vol. 20, no. 5, p. 1520, 2020.
- [42] A. Gupta et al., “Deep learning in image cytometry: a review,” *Cytometry Part A*, vol. 95, no. 4, pp. 366–380, 2019.
- [43] P. Baldi and P. Sadowski, “The dropout learning algorithm,” *Artif Intell*, vol. 210, pp. 78–122, 2014, doi: <https://doi.org/10.1016/j.artint.2014.02.004>.